



Basic Civil & Mechanical Engineering

R23 Regulation | JNTUK Kakinada

Year / Sem : 1st Year – 1st Semester (1-1)

Branch : Common for All Branches

Credits : 3 (Theory) + Lab Separate

Category : Engineering Science

### ■ Course Objectives

- Understand basic civil & mechanical engineering concepts
- Learn engineering fundamentals for real-world applications
- Develop practical knowledge of structures & machines

### ■ Course Outcomes

- Understand basic civil structures & construction materials
- Know machines, engines & thermal systems
- Apply engineering basics in practical workshop work

## UNIT – I : Basic Civil Engineering Concepts

### Importance of Civil Engineering

- Role of civil engineering in nation building & infrastructure
- Branches: Structural, Environmental, Geotechnical, Transportation

### Types of Buildings & Structures

- Residential, Commercial, Industrial & Special structures
- Load-bearing vs Framed structures

### Building Components

- Foundation – shallow (strip, raft) & deep (pile) foundations
- Superstructure – walls, columns, beams, slabs & roof
- Doors, Windows, Lintels, Arches and Staircases

### Introduction to Surveying

- Definition, objectives and types of surveying
- Chain surveying, levelling and contouring basics

## UNIT – II : Construction Materials

### Natural Materials

- Stones – classification, quarrying & properties
- Bricks – manufacturing process, types & quality tests
- Timber – seasoning, defects and uses

### Binding Materials

- Cement – types (OPC, PPC), composition & hydration process
- Lime – types and uses in construction

### Concrete & Steel

- Concrete – ingredients, water-cement ratio, workability
- Plain, Reinforced & Pre-stressed Concrete
- Steel – mild steel, TMT bars, structural steel sections

### Testing of Materials

- Slump test, compressive strength test for concrete
- Brick absorption test, cement fineness & soundness tests

## UNIT – III : Basic Mechanical Engineering

### Types of Machines & Mechanisms

- Simple machines – lever, pulley, screw, inclined plane
- Mechanical advantage, velocity ratio and efficiency

### Energy Sources

- Conventional – coal, petroleum, natural gas
- Non-conventional – solar, wind, hydro, nuclear energy

### Power Transmission Systems

- Belt & rope drives – open and cross belt
- Gear drives – spur, helical, bevel gears
- Chain drives and their applications

### Introduction to Engines

- Engine terminology – cylinder, piston, crankshaft, camshaft
- Working principle of 2-stroke and 4-stroke engines

## UNIT – IV : Thermal Engineering

### Thermodynamics Basics

- System, surroundings, properties – temperature, pressure, volume
- Zeroth Law – thermal equilibrium and temperature measurement

### Laws of Thermodynamics

- First Law – conservation of energy, heat and work interaction
- Second Law – Kelvin-Planck & Clausius statements
- Entropy concept and its significance

### Steam Boilers

- Classification – fire tube (Cochran, Lancashire) & water tube (Babcock)
- Boiler mountings and accessories

### IC Engines

- Petrol (SI) engine – Otto cycle working & P-V diagram
- Diesel (CI) engine – Diesel cycle & P-V diagram
- Engine performance – indicated power, brake power, efficiency

## UNIT – V : Manufacturing Processes

### Casting Process

- Pattern making – types of patterns & allowances
- Sand mould preparation & pouring, solidification defects

### Welding

- Arc welding (SMAW) & Gas welding (OAW) principles
- Weld joints – butt, lap, T-joint; welding defects

### Machining Processes

- Lathe operations – turning, facing, thread cutting, knurling
- Drilling, milling and grinding – working principle & uses

### Forming Processes

- Forging, rolling, extrusion & drawing – hot vs cold working
- Sheet metal operations – shearing, bending, deep drawing

### Workshop Practices

- Measuring tools – vernier caliper, micrometer, try square
- Workshop safety rules and personal protective equipment (PPE)

#### ■ BCME Lab / Workshop Practicals

- ◆ Carpentry – chiselling, planning, making joints
- ◆ Fitting – filing, marking, hacksaw, drilling
- ◆ Welding – arc welding lap & butt joints
- ◆ Sheet Metal – tray, funnel, cylinder making
- ◆ House Wiring – series, parallel, staircase circuits
- ◆ Workshop Safety – PPE, fire extinguisher usage

#### ■ Important Notes & Study Tips

- ◆ Foundation subject for all core engineering
- ◆ Draw clear diagrams for engines & structures
- ◆ Understand concepts – don't just memorise
- ◆ Important for GATE foundation & interviews
- ◆ Practice sketches of boilers, IC engines & BST